Model Monitoring Pipeline

Pipeline Stages

Telemetry Instrumentation

Record a request identifier, model version, audio duration, processing time, response status, and error category for each transcription. Avoid retaining audio unless it is needed and permitted for quality review. OpenTelemetry can standardize traces and metrics across the API and supporting services.

Operational Health Monitoring

Collect request volume, error rates, response times, and resource use such as CPU and memory. Prometheus can collect metrics and Grafana can display dashboards and alerts. CloudWatch is an option if the service runs on AWS. These tools help identify outages and performance problems, but they do not measure transcription accuracy.

Transcription Quality Evaluation

Regularly select a representative sample of transcriptions and have people compare them with reference transcripts. Calculate word error rate by counting substitutions, deletions, and insertions. Track the score over time and by relevant groups such as accent, audio length, and recording quality. Label Studio can help organize human review and labeling.

Input Data Drift Detection

Track characteristics such as sample rate, duration, volume, background noise, and the distribution of available accent or speaker metadata. Compare recent data with a baseline to spot changes that may affect accuracy. Evidently AI can help generate data quality and drift reports. A change in input data is a reason to investigate, not proof that model performance has worsened.

Performance Drift Investigation

When word error rate rises, examine the reviewed examples and input characteristics to understand the cause. Check for changes in users, recording conditions, audio processing, or

model version. LangChain is not a monitoring service for speech recognition. LangSmith is useful for tracing applications built with LangChain, but it does not replace accuracy evaluation for an ASR model.

Model Validation and Release

If a change in data or accuracy is confirmed, prepare representative examples and reference transcripts for evaluation. Compare a candidate model with the current model on the same test set. Check overall results and results across different accents and recording conditions to ensure improvements do not hide regressions.

Deploy an approved model gradually and monitor the same health and accuracy measures. Keep the previous model available so it can be restored if errors increase. Continue reviewing samples after release, document changes, and update baselines only when evidence shows that usage has changed consistently.